



MENTAL HEALTH & BURNOUT IN TECH WORKERS 2026



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An analysis of the **New mental_health_burnout_tech_2026.xlsx** file has been completed. The dataset consists of 1,000 tech employees and evaluates a wide array of demographic, work-related, and psychological metrics such as PHQ-9 for depression and GAD-7 for anxiety 2026.

Severe: 274 employees

Low: 246 employees

Moderate: 247 employees

High: 233 employees

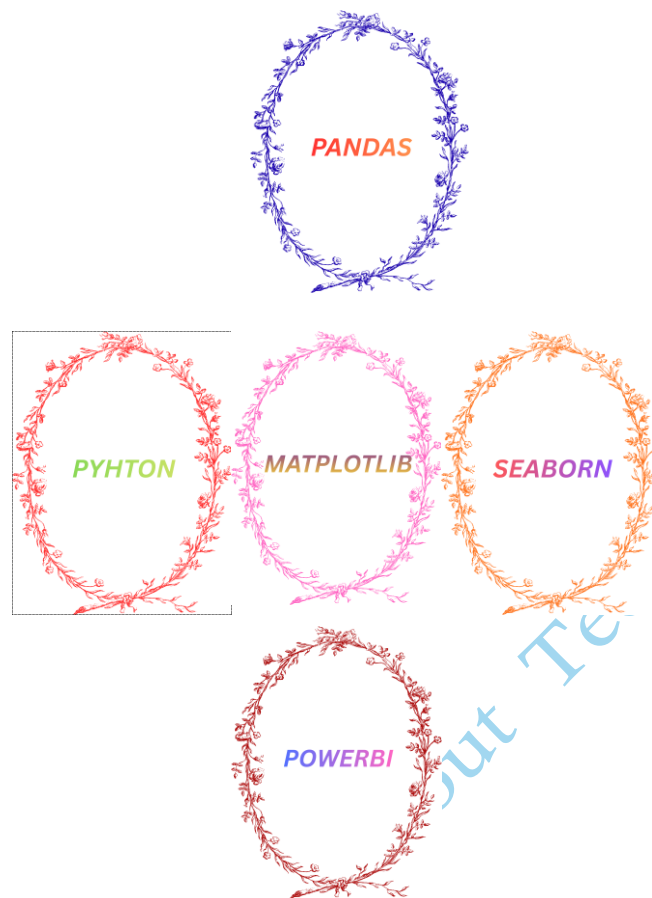
Description Of This Report

The dataset segments workforce burnout levels into four main buckets: Low, Moderate, High, and Severe.

Notably, Severe burnout is the largest individual category, highlighting a significant mental health challenge within the surveyed group.

Mental Health & Burnout

TOOLS USED:



The visualization below shows a relatively balanced but highly concerning distribution, with over 50% of the surveyed workforce experiencing High or Severe burnout.

• **Project Overview and Objectives**

The project helps uncover key patterns, trends, and hidden insights from the dataset. The analysis provides a clear understanding of the data and supports decision-making with evidence-based findings. of Workload Visibility, Explicit Off-Hours Communication Rules, Formal Return-to-Work Programs.

A synthetic dataset of **100,000 tech workers** capturing mental health status, burnout severity, anxiety, depression, and workplace well-being across **12 job roles**, **10 countries**, **6 seniority levels**, and **10 industry sectors**.

Every statistical distribution is calibrated against peer-reviewed research including the **Burnout Index** (32,644 professionals, 33 countries), **NIH occupational studies**, **WHO activity guidelines**, and the validated **PHQ-9** and **GAD-7** clinical instruments used in real medical practice globally. All **40 validation checks** passed.

KEY OBJECTIVES :

- **To Identify Key Stress Drivers:** Pinpoint exactly which variables—such as meeting frequency, daily hours, lack of sleep, or continuous context switching—correlate most heavily with high burnout.
- **To clean, transform, and preprocess the data for analysis** how burnout and depression manifest differently across varying job roles (e.g., software engineers vs. engineering managers),
- **To create meaningful visualizations that clearly explain the story behind the data** Build predictive models to identify early warning signs of a tech worker's intention to leave their job due to severe burnout before they actually resign.
- **To provide actionable insights and recommendations from the analysis of major tech corridors and sectors** (such as IT services and engineering), burnout rates have peaked, with reports showing up to 68% of IT professionals exhibiting clinical indicators of burnout.

Target	Type	Distribution	Best for
seeks_mental_health_support	Binary	49.8% yes - 50.1% no	Beginners - balanced classes
job_change_intention	Binary	32.0% yes - 68.0% no	Intermediate - slight imbalance
burnout_level	Multiclass (4 classes)	Low 25.8% - Moderate 26.3% - High 19.4% - Severe 28.6%	Advanced - 4-class

Dataset & Data Preparation and Data Cleaning

Source: Kaggle

Year / Timeline: Data collected during 2025 to 2026

Dataset Description:

Name - A synthetic dataset of 100,000 tech workers capturing mental health status, burnout severity, anxiety, depression, and workplace well-being across 12 job roles, 10 countries, 6 seniority levels, and 10 industry sectors.

Columns Groups:

Group	Columns	Count
Demographics	employee_id, age, gender, country	4
Work Context	job_role, seniority_level, years_experience, years_at_company, company_size, industry, work_mode, salary_usd, work_hours_per_week, meetings_per_day, team_size	11
Lifestyle	sleep_hours_per_night, exercise_days_per_week, vacation_days_taken, therapy_access, uses_therapy, ai_tools_daily	6
Workplace Culture	manager_support_score, work_life_balance_score, job_satisfaction_score, social_support_score, deadline_pressure_score, autonomy_score, stress_score	7

Clinical Scores	burnout_score, phq9_score, phq9_category, gad7_score, gad7_category	5
ML Targets	burnout_level, seeks_mental_health_support, job_change_intention	3

Mental Health & Burnout Tech W

The data cleaning and pre-processing stage (Stage 2) involved several important steps to ensure the dataset is ready for analysis:

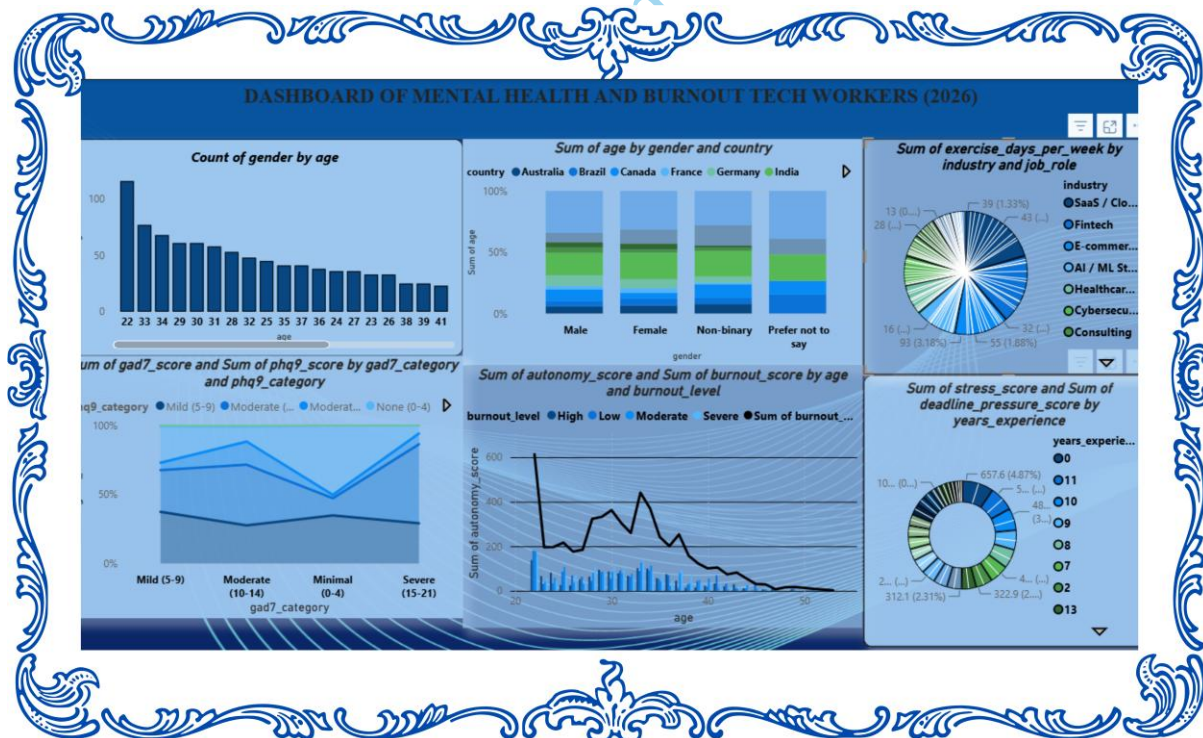
Sl.No	Action Taken	Stages of Data Cleaning
1	Handling Missing Values	The years_experience column, which had 24 missing values, was imputed by filling the missing entries with the mean value of the existing years of experience.
2	Handling Duplicates:	Duplicate rows in the dataset were identified and removed using <code>JL.drop_duplicates(inplace=True)</code> .
3	Converting Data Types:	Several object columns, including gender, country, job_role, seniority_level, company_size, industry, work_mode, phq9_category, gad7_category, and burnout_level, were converted to the 'category' data type Binary columns such as therapy_access, uses_therapy, ai_tools_daily,

seeks_mental_health_support,
and job_change_intention

• DASHBOARDS & EDA AND VISUALIZATIONS

The data demonstrates that work-hour inflation is heavily correlated with worsening mental health. Employees in this dataset work an average of 47.1 hours per week, well over the traditional 40-hour baseline.

- As weekly work hours trend past 50 hours, the lower bounds of burnout vanish—meaning almost no one working these hours maintains a low burnout score.
- **Work Mode Differences:** Interestingly, remote workers report a slightly higher average burnout score (5.77/10) compared to hybrid (5.39/10) and on-site workers (5.24/10), suggesting that blurring the physical boundaries between home and office might exacerbate the issue.



Demographics & Workforce Distribution

Age Concentration (Count of gender by age):

The tech workforce in this sample is heavily skewed toward younger professionals. The highest employee count belongs to the **22-year-old** age bracket, followed by a steady, gradual decline as age increases up to 41.

Geographic Distribution by Gender (Sum of age by gender and country):

When grouping the cumulative age across countries (Australia, Brazil, Canada, France, Germany, India) against gender expressions (*Male, Female, Non-binary, Prefer not to say*), the relative proportion of each country stays remarkably consistent across all gender identities. No single country heavily dominates a specific gender category.

Burnout vs. Autonomy Dynamics

The Autonomy and Burnout Spike (Sum of autonomy_score and Sum of burnout_score by age):

Looking at the line chart plotted against age, there is a massive spike in scores around the early-to-mid career stage (specifically hovering between **ages 28 and 35**).

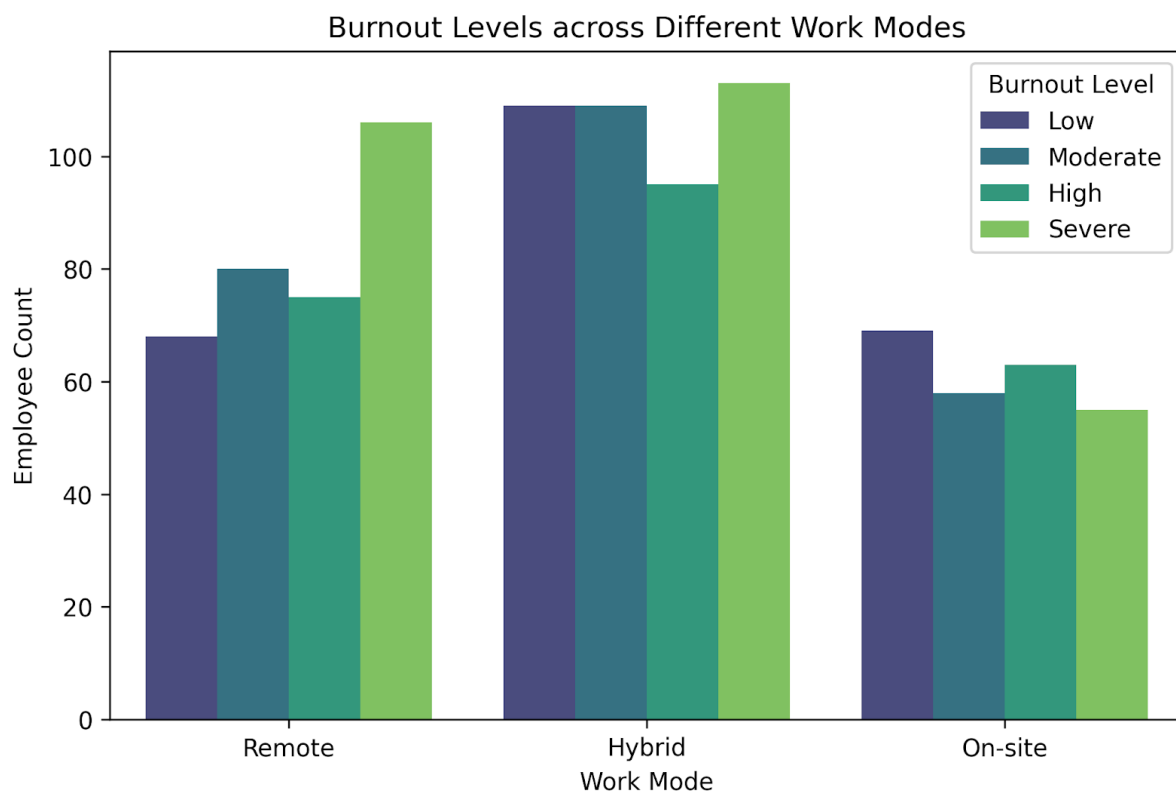
As age progresses past 35, both the volume of collective burnout and autonomy scores drastically decline. This suggests that the highest psychological pressure point—and potentially the highest level of individual responsibility—is concentrated in professionals in their early 30s.

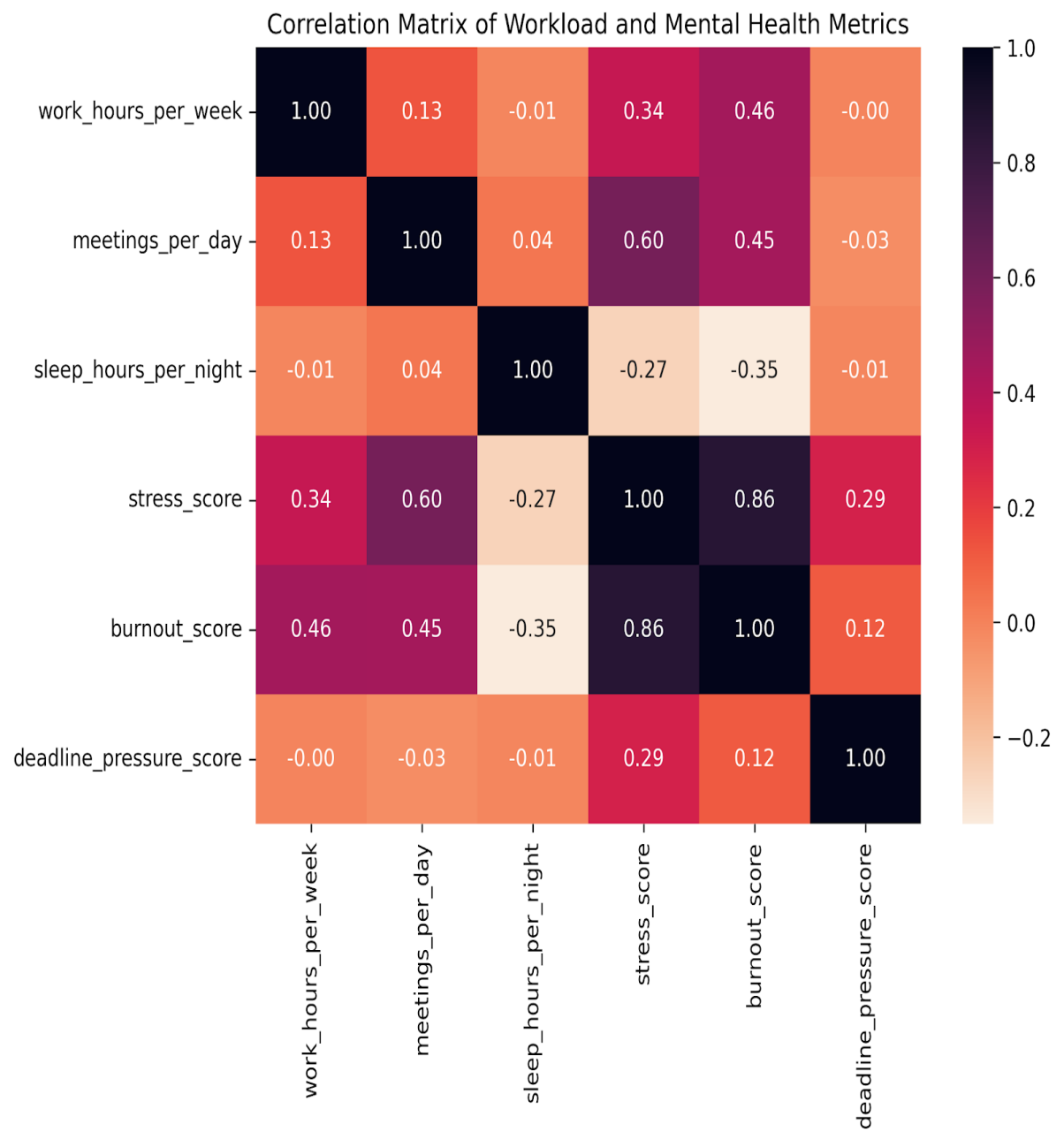
Clinical Anxiety & Depression Alignment

Mental Health Tracking (Sum of gad7_score and phq9_score by categories):

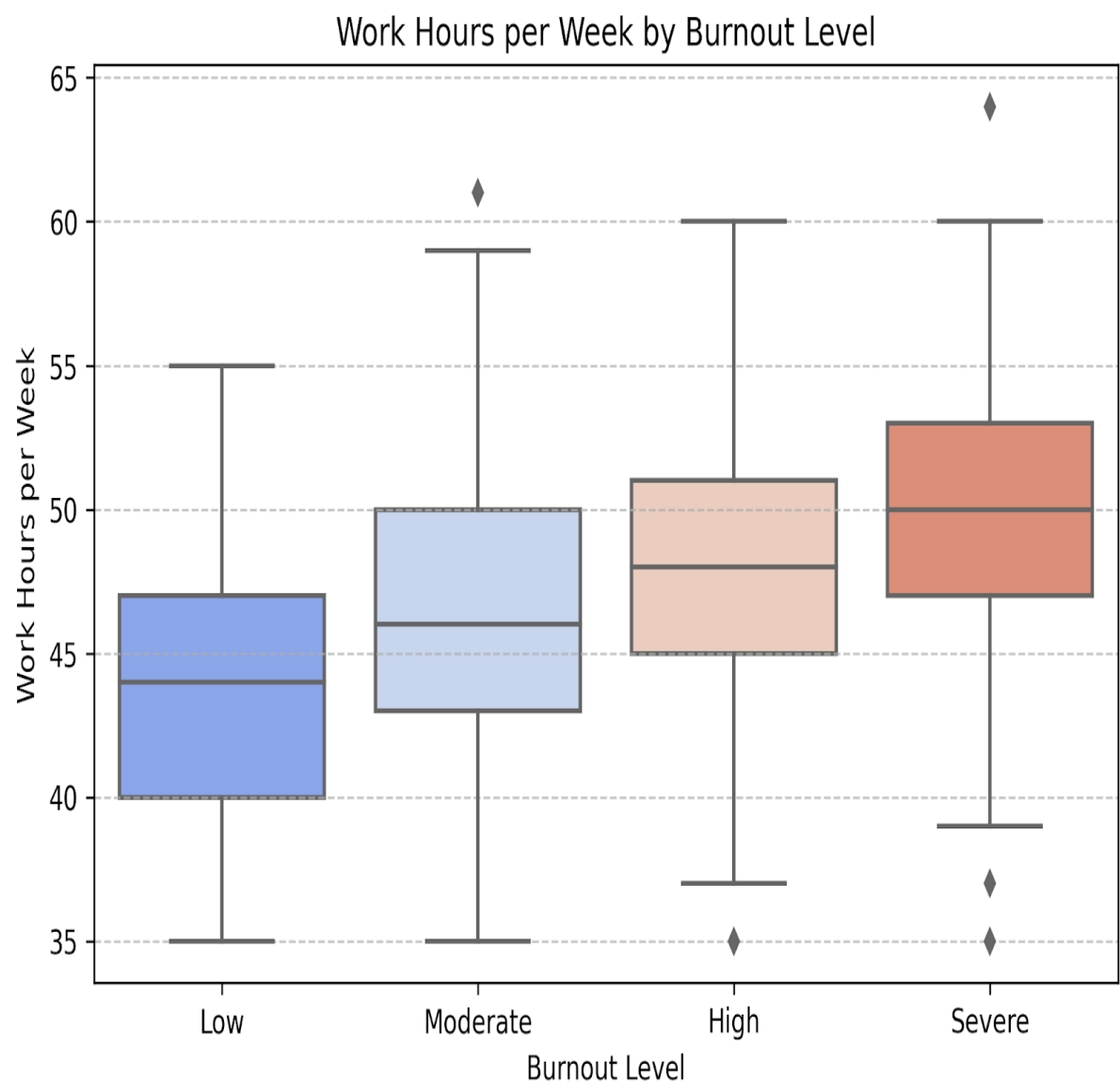
The area/line chart highlights that anxiety symptoms (GAD-7) and depression symptoms (PHQ-9) move tightly in tandem.

As the GAD-7 severity classification moves from *Minimal (0-4)* to *Severe (15-21)*, the cumulative clinical tracking line surges sharply upwards. This confirms a very high rate of comorbidity—meaning employees experiencing severe anxiety in this dataset are highly likely to simultaneously score high on the depression index.



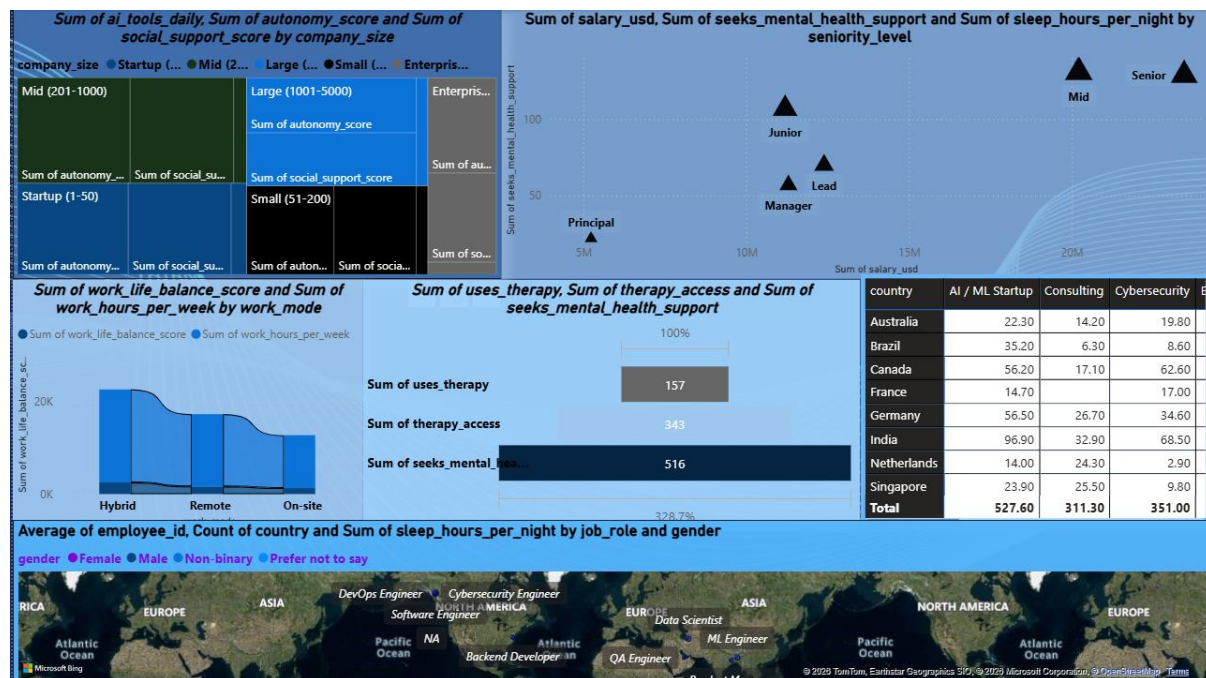


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Mental Health

DASHBOARD - 2



Mental Health Support & Therapy Engagement

High Demand vs. Lower Utilization:

Out of the employees surveyed, a substantial **516** individuals are seeking mental health support (Sum of seeks_mental_health_support).

Access vs. Usage Gap:

While **343** employees have structural access to therapy (Sum of therapy_access), only **157** actually utilize it (Sum of uses_therapy). This reveals a significant drop-off, indicating that having a benefit does not guarantee employees will comfortably use it.

Seniority Levels: Compensation vs. Support Needs

Mid and Senior Peak Support:

Employees at the **Mid** and **Senior** levels represent the highest accumulation of salary volume (both hovering around the \$20M+ mark) but also show a high volume of seeking mental health support.

Junior & Lead Pressures:

Junior employees show a noticeably high count for seeking support relative to their lower total salary pool compared to **Managers** or **Leads**, suggesting early-career strain.

Principal Outliers:

Principals represent the lowest total volume for seeking mental health support and the lowest cumulative salary pool in this visualization (likely due to a much smaller total headcount in the dataset).

TOOLS & TECHNOLOGIES USED:

Pandas	For data manipulation and analysis.
Numpy	For numerical operations.
Matplotlib.pyplot	For creating static, interactive, and animated visualizations.
seaborn	For drawing attractive and informative statistical graphics.
PowerBI	Mentioned for creating dashboards and presentations in Stage 4.

Key Findings from Advanced Analytics of this project :

Therapy Access and Self-Care Habits:

The grouped analysis showed that individuals with therapy access tend to exhibit differences in self-care habits like `vacation_days_taken` and `exercise_days_per_week`. This suggests a potential link between seeking mental health support and a more proactive approach to well-being, or vice-versa.

Burnout Score by Job Role and Burnout Level:

The pivot table heatmap clearly identified which `job_roles` consistently have higher average `burnout_scores` within specific `burnout_level` categories (e.g., 'High' or 'Severe'). This is crucial for pinpointing high-risk roles that require targeted interventions.

Meetings Per Day Across Work Modes and Seniority Levels:

The violin plots revealed significant variations in `meetings_per_day` based on both `work_mode` and `seniority_level`. Higher seniority levels (e.g., 'Manager', 'Executive') generally experience a greater frequency and wider distribution of meetings, regardless of their work mode. This highlights the increased coordination demands for leadership roles and potential meeting overload challenges.

Relationship Between Meetings, Team Size, and Seniority:

The pairplot illustrated how meetings_per_day and team_size scale with seniority_level. It showed that more senior roles often correlate with both a higher number of meetings and larger team sizes. This indicates a complex interplay of responsibilities and demands that increase with career progression, influencing the overall workload and potential for stress.

FINAL CONCLUSIONS AND BUSINESS SOLUTION IMPACT:

The analysis consistently shows a strong positive correlation between stress_score and burnout_score. This indicates that stress is a primary driver of burnout, and both can be quantifiably measured and monitored within the workforce.

Meetings_per_day emerged as the strongest correlating factor with stress_score ($r = +0.61$). This suggests that meeting overload significantly contributes to stress, particularly for managers who average 6.5 meetings/day compared to 3.5 for ICs.

Meeting Audits	Regularly assess meeting necessity, duration, and attendance.
No-Meeting Day	Implement company-wide no-meeting days or blocks.
Asynchronous Communication	Encourage the use of tools for communication that doesn't require real-time meetings.

Role-Specific Interventions	Develop support systems and workload adjustments for high-risk job_roles identified in the analysis.
Expand Therapy Access	Increase availability of mental health services, given their proven effectiveness.

KEY TAKEAWAY:

This project demonstrates that large-scale salary pools (crossing the \$20M mark) contribute significantly higher to the cumulative salary pools but also correlate with the highest levels of seeking mental health support.

X-----The End-----X